

LineDraw: Converting Spectrograms into Sparse Musical Tokens

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Abstract

We propose LineDraw, a signal-processing-based framework that represents audio as a collection of sparse line events in the time-frequency domain. Unlike neural approaches such as DDSP, LineDraw does not rely on learned parameters. Instead, it extracts continuous frequency trajectories from normalized multi-window CQT representations and reconstructs audio through sinusoidal synthesis. Experimental results demonstrate that the extracted line representation preserves both musical note information and spectral structure. We further discuss its potential applications in music language modeling, audio understanding, and large-scale self-supervised music learning.

1 Introduction

Audio is highly structured, and many musical events appear as continuous trajectories in the time-frequency domain. Current neural approaches often model audio as dense tensors. We investigate whether a sparse line representation can provide a simpler and more interpretable alternative.

2 Related Work

2.1 Sinusoidal Modeling

2.2 Spectral Modeling Synthesis

2.3 DDSP and Neural Audio Synthesis

DDSP: audio = neural network + sinusoidal/noise synthesizer

LineDraw: audio = signal processing + line extraction + sinusoidal/noise synthesizer

2.4 AMT and Sparse Coding

3 Method

3.1 Multi-window CQT

Given an audio signal $x(t)$, we compute a multi-window CQT stack:

$$X_k(f, \tau) = \text{CQT}_{w_k}\{x\}(f, \tau), \quad (1)$$

where w_k denotes the k -th analysis window.

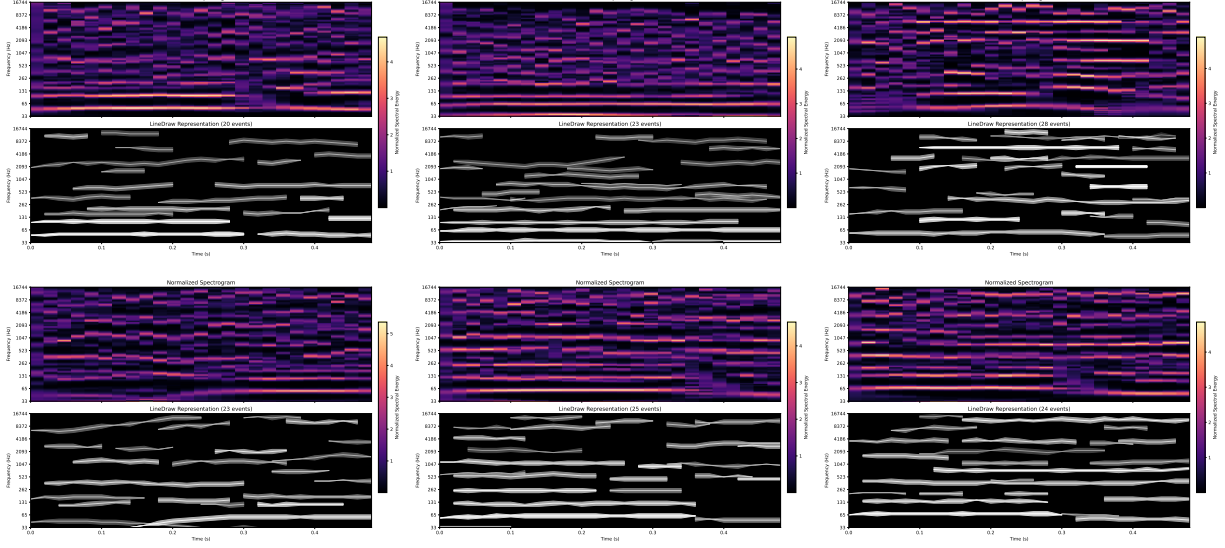


Figure 1: Additional examples of LineDraw representations. Each example shows the normalized spectrogram (top) and the corresponding extracted line events (bottom). The examples are randomly sampled from the evaluation dataset.

3.2 Local Frequency Normalization

We normalize local energy to reduce amplitude bias:

$$\tilde{X}(f, \tau) = \frac{X(f, \tau)}{\epsilon + \mathcal{N}(X)(f, \tau)}, \quad (2)$$

where $\mathcal{N}(\cdot)$ is a local neighborhood statistic and $\epsilon > 0$ is a stability constant.

3.3 Line Extraction

Line candidates are extracted by thresholding and continuity constraints in the normalized CQT plane. A line event is defined as a connected trajectory

$$\ell_i = \{(f_t, t)\}_{t=t_s}^{t_e}, \quad (3)$$

with minimum duration and smoothness constraints.

3.4 Line Event Representation

Each line is represented as a tuple:

$$e_i = (t_s, t_e, f(t), a(t), \phi_0), \quad (4)$$

where $f(t)$ is instantaneous frequency, $a(t)$ is amplitude envelope, and ϕ_0 is initial phase.

3.5 Sinusoidal Resynthesis

Reconstruction is obtained by summing all extracted line events:

$$\hat{x}(t) = \sum_i a_i(t) \cos\left(2\pi \int_0^t f_i(\tau) d\tau + \phi_i\right). \quad (5)$$

4 Experiments

4.1 Dataset Construction

To evaluate the proposed LineDraw representation, we constructed a manually annotated pitch dataset from YouTube recordings.

Audio clips were collected from a diverse set of musical sources retrieved using YouTube search. We queried multiple keywords spanning different musical styles, including popular music, piano performances, electronic dance music, jazz, orchestral music, rhythm games, and independent composers. The retrieved videos were randomly sampled and segmented into short audio excerpts.

Each sample consists of a 0.5-second stereo audio segment sampled at 44.1 kHz. During annotation, a Constant-Q Transform (CQT) visualization was provided as an auxiliary tool to facilitate pitch labeling. Annotators were asked to identify all salient pitches present in each segment. The resulting dataset contains 1,155 annotated audio clips.

Unlike conventional Automatic Music Transcription (AMT) datasets, the proposed dataset only contains clip-level pitch annotations and does not provide note onset or offset information. Consequently, evaluation focuses on pitch presence rather than temporal note alignment.

Figure 2 summarizes the statistical properties of the dataset. The event-count distribution illustrates the number of annotated pitches per clip. The pitch-class histogram demonstrates a relatively balanced distribution among the twelve pitch classes, while the octave histogram indicates that most annotated notes concentrate in the middle musical register.

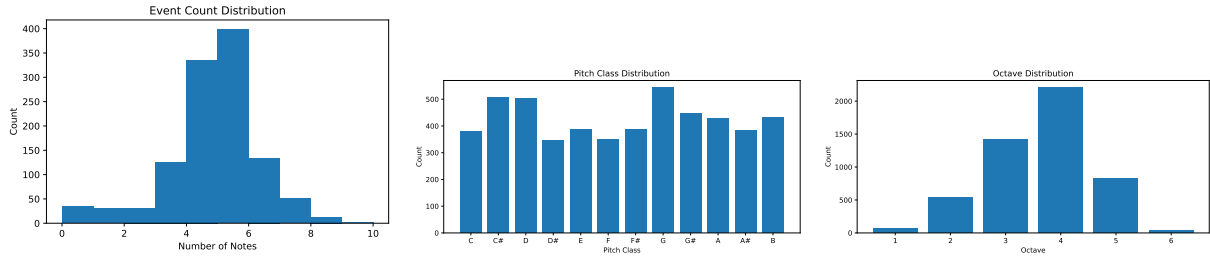


Figure 2: Statistical analysis of the collected pitch dataset. Left: distribution of the number of annotated pitches per audio segment. Middle: pitch-class distribution. Right: octave distribution. The dataset contains 1,155 manually annotated 0.5-second audio clips collected from diverse YouTube sources.

4.2 Note Recovery

Metrics: Precision, Recall, F1.

Ground truth: MIDI note annotations.

Prediction: LineDraw note list mapped from extracted line events.

4.3 Spectral Reconstruction

Metrics: L1 distance and MSE between reference and reconstructed spectrograms.

4.4 Threshold Analysis

Metrics: ROC curve and AUC under varying extraction thresholds.

Table 1: Performance of LineDraw under different extraction thresholds. ECR denotes the Explained Energy Ratio, and Events denotes the average number of extracted line events per audio clip.

Threshold	MIDI-Level			Pitch-Class			ECR $\uparrow$	Events $\downarrow$
	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$		
0.7	0.116	0.794	0.203	0.273	0.990	0.428	0.892	76.75
1.0	0.140	0.739	0.236	0.290	0.967	0.446	0.813	55.86
2.0	0.233	0.559	0.329	0.404	0.837	0.545	0.543	20.55
3.0 Best	0.304	0.395	0.343	0.526	0.652	0.582	0.328	8.19
4.0	0.342	0.213	0.263	0.631	0.415	0.501	0.182	3.19
4.3	0.348	0.157	0.216	0.663	0.331	0.441	0.142	2.23

5 Discussion

5.1 Interpretability

The line representation offers explicit event-level trajectories that are easier to inspect than dense latent tensors.

5.2 Computational Cost

5.3 Failure Cases

6 Future Work

6.1 Line Transformer

Learning sequence models directly over line events.

6.2 Audio LLM

Using line tokens as a compact interface for multimodal audio-language reasoning.

6.3 Large-scale Self-Supervised Learning

Pretraining with masked line modeling and trajectory prediction.

6.4 Music Generation

Controllable synthesis by editing line event sequences.

7 Conclusion

LineDraw provides a sparse and interpretable audio representation based on time-frequency line extraction, with competitive note and spectral reconstruction quality and promising downstream potential.